SAE Faithfulness: Project Summary

GPT-2 Small (frozen in every experiment; only SAE parameters are trained) · SAEs at `blocks.L.hook_resid_pre`, BOS excluded · Evidence labels: **TEST** = single pre-registered M5 run; **VAL** = validation · Prepared 2026-11-10 from committed result files.
Which results are shown: results from before Nov 3 are left out. "Old" always means the **corrected post-review baseline** (one uninterrupted training run per layer, 3 seeds, matched L0).

> **Research question.** Can we improve SAE reconstruction of frozen LLM activations while preserving truth-related information and causal behaviour across transformer layers?
> **Answer.** Reconstruction and behaviour improved substantially and were confirmed on test. Truth information did not improve, and SAEs never beat PCA. Causal preservation improved slightly at L8 (validation) and got worse after KL fine-tuning at L11. **The four faithfulness levels dissociate.**

Part A. Experiment guide

R-series (independent review, Nov 10): improvement experiments

ID	Name	What it did	Result	Status
R1	Public SAE reference (= E2)	Ran the public res-jb SAE (about 300M training tokens, 24,576 latents) through our full evaluation	Our conclusions also hold for a strong external SAE	VAL descriptive
R2	Data-scaled SAEs	Retrained on about 100M tokens at 4× width with a modern recipe; L0 = 128; layers 1/8/11; 3 seeds	NMSE −31 to −74 %; ΔCE −35 to −79 %; truth unchanged	TEST confirmed
R3	Gemma-2-2B + Gemma Scope SAEs	Planned cross-model check	—	NOT RUN
R4	KL fine-tune at fixed L0	Fine-tuned R2 TopK to keep GPT-2's predictions unchanged, against an MSE-only control	ΔCE −16 to −17 %; NMSE +2 %; causal worse at L11	TEST behaviour; VAL causal
R5	Causal skyline	Compared SAE causal preservation with learned (DAS), PCA and random low-rank subspaces	A rank-4 DAS subspace carries more than the whole SAE at L8	VAL descriptive

> **Naming clash.** "R1/R2" was also used on Nov 7–8 for two RepE honesty experiments: an honesty-preservation fine-tune and an SAE feature clamp. Both failed and are now marked HISTORICAL (see Part D).

M-series (measurement framework, Nov 8–10)

ID	What it is	Key result
M0	Re-analysis of 107 matched SAEs: within one layer, does reconstruction error predict behaviour and truth?	Error predicts ΔCE (Holm p = 0.0008) but **not** the truth gap
M1	Built **RepE-LM**, a confound-free persona task (honest and dishonest prompts differ by exactly one token)	All 7 quality gates pass; the effect is real but small (median KL 0.019 nats)
M2	Four-level information readers: frozen probe, noise reference, re-fit probe, PCA reference	The GoT frozen-probe collapse is mostly probe fragility; **SAE never beats PCA**
M3	Where GPT-2 carries the persona effect (activation patching, 65 cells)	Persona slot early (97 % at L1), final token late (95 % at L11); handover at L7–8
M4	How much of that causal effect passes through the SAE	93–97 % at L1; **62–68 %** at L8/L11
M5	Final pre-registered TEST run: 81 SAEs, 552 test rows, 136 comparisons	Confirms R2/R4 on OWT; **0/34** GoT improvements

Other experiments:

- **E1** (TopK matched on task-prompt L0): **not run**.

- **E3** (do fine-tunes help causal preservation?): no.
- **IMP-L / RLM-RP** (multi-layer fine-tunes): no creditable truth or persona gain.

Part B. OpenWebText: reconstruction and behaviour

B1. Pipeline

```
OWT text → GPT-2 tokenizer (≤128 tokens, BOS excluded)
  → FROZEN GPT-2 Small → residual x at blocks.L.hook_resid_pre (768-d, every token)
  → SAE encoder → sparse z (≈128 active of 3,072 or 12,288) → decoder → x̂
  → (1) RECONSTRUCTION: compare x̂ with x
  → (2) SPLICING: replace x with x̂ inside GPT-2, run layers L..11, compare next-token predictions
       ΔCE = CE(spliced) − CE(original);  KL = mean KL(p_orig ‖ p_spliced)
```

B2. Evaluation types and metrics

Type	Metric	Better	Meaning
Reconstruction	NMSE = 1 − EV	↓	Share of activation variance lost
	EV (explained variance)	↑	Share of variance kept
	Cosine(x, x̂)	↑	Direction kept
	L0	= 128	Active latents per token (kept matched within ±10 %)
	Dead fraction	↓	Latents that never fire
Behaviour (splicing)	ΔCE	↓	Extra next-token loss in nats/token (original test CE 3.30)
	KL(p_orig ‖ p_spliced)	↓	How much GPT-2's prediction distribution changes

B3. Layers used

Phase	Experiment	Layers	Split
Corrected baseline	Corrected protocol (108 SAEs) + splicing	**all 12**	test
Nov 10	R2, E2, M5 (R4: 8 and 11 only)	**1, 8, 11**	val + test

B4. Problems found in the review and fixes

Problem	Fix	Effect
Different optimiser procedure per layer (L6 resumed with Adam restored; L2/L10 warm-started with Adam reset)	One uninterrupted 15k-step run for every layer	A validity fix; architecture rankings unchanged
Single seed	3 seeds (42 / 1042 / 2042)	Error bars available
Only layers 2/6/10	All 12 layers (108 SAEs)	Full depth profile
Runs with unmatched L0 averaged in the architecture summary	±10 % L0 rule; summary declared INVALID	107/108 SAEs matched
No behavioural metric	Splicing (ΔCE, KL)	Measures what GPT-2 actually loses
Dead features not tracked	Logged	Old L1: TopK 44 % dead, JumpReLU 68 % dead
Under-trained SAEs (101,600 tokens, about 38 passes)	**R2** (below)	See B6

B5. Corrected baseline at layers 2/6/10 (OWT test, 12,700 tokens, width 3,072, 3-seed mean ± sd)

Layer	Arch	MSE	NMSE (1 − EV)	EV	Cosine	L0
2	ReLU	0.1123 ± 0.0006	0.0642	0.9358 ± 0.0003	0.9887	131.9
2	TopK	0.1296 ± 0.0006	0.0741	0.9259 ± 0.0003	0.9867	128.0
2	JumpReLU	0.1128 ± 0.0030	0.0645	0.9355 ± 0.0017	0.9877	129.0
6	ReLU	0.7580 ± 0.0037	0.1685	0.8315 ± 0.0008	0.9561	131.0
6	TopK	0.6422 ± 0.0011	0.1428	0.8572 ± 0.0002	0.9621	128.0
6	JumpReLU	0.6076 ± 0.0020	0.1351	0.8649 ± 0.0004	0.9627	129.1
10	ReLU	3.6097 ± 0.0062	0.2273	0.7727 ± 0.0004	0.9401	132.7
10	TopK	2.8195 ± 0.0067	0.1775	0.8225 ± 0.0004	0.9509	128.0
10	JumpReLU	2.8162 ± 0.0042	0.1773	0.8227 ± 0.0003	0.9510	131.6

- **MSE grows with depth** (0.11 at L2 to about 2.8–3.6 at L10) because the residual stream gets larger in later layers. Compare layers with NMSE/EV, not raw MSE.
- R2 SAEs were trained only at layers 1/8/11, so there are no R2 values for layers 2/6/10.

B6. What R2 changed

	Old	R2
Training data	101,600 tokens × about 38 passes	about 100M fresh tokens, single pass (overlapping documents removed)
Width	3,072	12,288 (plus a 3,072 ablation)
Dead-latent fix	none	AuxK (k_aux 512, weight 1/32)
Optimiser	Adam 3e-4 constant, batch 256, clip 10	Adam 2e-4, warm-up + linear decay, batch 4,096, clip 1
Other	—	Input scaling ($E\|x\|^2 = 768$), tied initialisation, decoder bias = data mean
Architectures	ReLU+L1, TopK, JumpReLU (sigmoid surrogate)	ReLU (decoder-norm L1), TopK+AuxK, **BatchTopK**+AuxK (modern JumpReLU would not settle at L0 128)

B7. Old (corrected baseline) vs R2: OWT TEST (M5), 3-seed mean, L0 ≈ 128 for all

Layer	Arch (old → R2)	NMSE	EV	Cosine	ΔCE (nats)	KL
1	ReLU	0.063 → 0.023 **−64 %**	0.937 → 0.978	0.990 → 0.996	0.0141 → 0.0038 **−73 %**	0.0199 → 0.0054 **−73 %**
1	TopK	0.079 → 0.025 **−69 %**	0.921 → 0.975	0.987 → 0.996	0.0221 → 0.0056 **−75 %**	0.0259 → 0.0063 **−76 %**
1	JumpReLU → BatchTopK	0.068 → 0.018 **−74 %**	0.933 → 0.982	0.988 → 0.997	0.0088 → 0.0019 **−79 %**	0.0139 → 0.0034 **−76 %**
8	ReLU	0.195 → 0.132 **−32 %**	0.805 → 0.868	0.946 → 0.963	0.218 → 0.131 **−40 %**	0.236 → 0.140 **−41 %**
8	TopK	0.160 → 0.094 **−41 %**	0.840 → 0.906	0.954 → 0.973	0.133 → 0.062 **−53 %**	0.140 → 0.066 **−53 %**
8	JumpReLU → BatchTopK	0.156 → 0.090 **−43 %**	0.844 → 0.910	0.955 → 0.974	0.121 → 0.054 **−55 %**	0.131 → 0.059 **−55 %**
11	ReLU	0.243 → 0.168 **−31 %**	0.757 → 0.832	0.953 → 0.966	0.342 → 0.221 **−36 %**	0.349 → 0.223 **−36 %**
11	TopK	0.185 → 0.111 **−40 %**	0.815 → 0.889	0.962 → 0.977	0.204 → 0.096 **−53 %**	0.206 → 0.097 **−53 %**
11	JumpReLU → BatchTopK	0.185 → 0.107 **−42 %**	0.815 → 0.893	0.962 → 0.978	0.216 → 0.093 **−57 %**	0.218 → 0.092 **−58 %**

Robustness:

- Every cell improved on 9/9 seed pairs, with the Welch 95 % CI excluding 0.
- Dead latents fell from 44–68 % (old, L1) to about 0 %.
- JumpReLU → BatchTopK compares different architectures.

R4 on top of R2 (TopK, OWT TEST): KL fine-tune vs MSE-only control on the same 5M tokens			
Layer	**ΔCE: MSE control → KL**	**Measured KL: MSE → KL**	**NMSE: MSE → KL**
8	0.0623 → 0.0514 **−17.4 %** (3/3 seeds)	0.0659 → 0.0564 **−14.4 %**	0.0938 → 0.0958 **+2.1 %**
11	0.0964 → 0.0807 **−16.3 %** (3/3 seeds)	0.0968 → 0.0813 **−16.0 %**	0.1109 → 0.1133 **+2.2 %**

Where the R2 gain came from (TopK, TEST, % change)			
Layer	**A1: data + recipe (width 3,072): NMSE / ΔCE**	**A2: width 3,072 → 12,288: NMSE / ΔCE**	**Total: NMSE / ΔCE**
1	−35.0 % / −40.3 %	−51.6 % / −58.0 %	−68.6 % / −74.9 %
8	−22.0 % / −36.6 %	−24.5 % / −26.5 %	−41.1 % / −53.4 %
11	−24.6 % / −33.6 %	−20.2 % / −29.1 %	−39.8 % / −52.9 %

> A1 changes token count **and** recipe together, so the gain cannot be credited to more data alone. The 10M/25M/50M checkpoints exist but were not evaluated.

Part C. Geometry of Truth: truth information

C1. Pipeline

```
"The city of Krasnodar is in Russia."  → TRUE    "… is in South Africa." → FALSE   (cities + neg_cities)
  → split grouped by CITY: train 2,092 / validation 448 / test 452  (no city appears in two splits)
  → FROZEN GPT-2 → residual at blocks.L.hook_resid_pre, FINAL token → x (768-d)
  → ORIGINAL truth probe: StandardScaler + logistic regression on x (train only; C by city-grouped CV)
  → SAE (trained on OWT only) → x̂ → scored by four readers:
       1. FROZEN probe  — original probe applied to x̂ unchanged
       2. NOISE         — x + random noise of the SAE error's norm (200 draws), same frozen probe
       3. RE-FIT probe  — new probe trained on x̂
       4. PCA           — PCA with the same OWT error as the SAE, then a re-fit probe
  → 3 SAE seeds + 1,000 city bootstraps; M5 rule: all 9 seed pairs better AND 95 % CI excludes 0
```

C2. Metrics (all gaps: lower is better)

Metric	Formula	Question it answers
Original AUROC	AUROC of the probe on x	Is truth readable at this layer? (informative if ≥ 0.70; L8 0.84, L11 0.79 on test)
Frozen gap	$\text{AUROC}(x) - \text{AUROC}_{\text{frozen}}(\hat{x})$	Does the model's own truth readout survive the SAE?
Noise gap	$\text{AUROC}(x) - \text{AUROC}_{\text{frozen}}(x + \text{noise})$	How fragile is the probe to *any* error of this size?
Re-fit gap	$\text{AUROC}(x) - \text{AUROC}_{\text{refit}}(\hat{x})$	Is the truth *information* still in x̂?
SAE − PCA	re-fit gap(SAE) − re-fit gap(PCA)	Does the SAE keep truth better than PCA at equal error? (< 0 = yes)

C3. Problems found in the review and fixes

Problem	Fix	Corrected result
The L10 "faithfulness" probe was re-trained on x̂ (measures decodability, not the model's readout)	Frozen original probe	res-jb L10 frozen gap **0.214** [0.16, 0.27]
Truth-aware fine-tune was circular (evaluation probe = training teacher)	Independent probes (train groups A/B)	MLP gap 0.189, **worse** than recon-only's 0.107: method abandoned
Fine-tune L0 drift	±5 % L0 rule	Earlier fine-tune gains withdrawn
No noise or PCA reference	M2 four readers	Most of the frozen gap is probe fragility
Test scored repeatedly; single seed	Single pre-registered M5; 3 seeds	Clean confirmation

C4. Corrected SAEs, all four readers (VAL; ReLU / TopK / JumpReLU, 3-seed mean)

Layer	Orig. AUROC	Frozen gap	Noise gap	Re-fit gap	SAE − PCA
6	0.841	0.320 / 0.287 / 0.318	0.300 / 0.291 / 0.299	0.067 / 0.080 / 0.141	+0.017 / +0.037 / +0.101
8	0.844	0.272 / 0.287 / 0.276	0.266 / 0.252 / 0.263	0.034 / 0.061 / 0.085	+0.019 / +0.045 / +0.070
10	0.852	0.286 / 0.271 / 0.299	0.253 / 0.234 / 0.249	0.065 / 0.090 / 0.114	+0.024 / +0.063 / +0.089
11	0.825	0.270 / 0.250 / 0.249	0.234 / 0.217 / 0.226	0.105 / 0.110 / 0.126	+0.042 / +0.072 / +0.087

- Noise of the same size causes almost the whole frozen gap, so the probe is fragile.
- The real information loss (re-fit gap) is 0.03–0.14.
- The SAE is never better than PCA.

C5. Old vs R2: GoT TEST (M5), 3-seed mean (R2 order: ReLU / TopK / BatchTopK)

Layer	Metric	Old SAEs	R2	R4 KL	Verdict
8	frozen gap	0.275 / 0.276 / 0.280	0.281 / 0.247 / 0.307	0.241	no clear change
8	re-fit gap	0.034 / 0.061 / 0.058	0.028 / 0.056 / 0.065	0.048	no clear change
8	SAE − PCA	+0.008 / +0.036 / +0.034	+0.010 / +0.042 / +0.053	+0.035	**never beats PCA**
11	frozen gap	0.221 / 0.228 / 0.210	0.237 / 0.197 / 0.186	0.205	no clear change
11	re-fit gap	0.077 / 0.101 / 0.107	0.052 / 0.077 / 0.087	0.078	trend −19 to −32 %, **not significant**
11	SAE − PCA	+0.040 / +0.072 / +0.079	+0.030 / +0.072 / +0.081	+0.073	**never beats PCA**

- **0 of 34 GoT test cells improved.** Two regressed: extra width at L8 (re-fit gap +0.012 [+0.004, +0.020]).
- A random-decoder SAE ties PCA (SAE − PCA −0.003 at L8, −0.013 at L11; n = 1).
- ReLU's better re-fit gap is confounded: it uses fewer latents on GoT prompts (77–112 vs 128).

Part D. RepE: persona and honesty

D1. Pipeline

```
RepE-LM task (built in M1):
  506 statements × 20 variants = 10,120 prompts; one-token swap (truthful/deceptive, sincere/lying, …)
  split by statement: train 417 / validation 89   (no test split was built)
  → FROZEN GPT-2 → residual at the persona SLOT (L1) or the FINAL token (L8, L11) → SAE → x̂
  → (a) INFORMATION: frozen / noise / re-fit / PCA probes for honest vs dishonest
    (b) CAUSAL: patch the honest−dishonest difference Δ through the SAE and measure the effect on
        GPT-2's next-token distribution: T(v) = KL(p_H‖p_U) − KL(p_H‖p_U+v);  T_rec/T_full
```

D2. Metrics

Type	Metric	Better
Information	Frozen gap, noise gap, re-fit gap, SAE − PCA (as for GoT); original persona AUROC L8 0.85, L11 0.87	↓
Causal	T_rec/T_full = share of the persona effect carried by the SAE reconstruction	↑ (1 = all)
Skyline (R5)	Share carried by a rank-k DAS / PCA / random subspace	reference

D3. Problems found in the review and fixes

Problem	Fix
Length confound: in the original honest/untruthful prompts, " untruthful" is longer than " honest", so prompt length alone predicted the label	**M1** RepE-LM: one-token swap; every shortcut at AUROC 0.500
The original "steering" test had no SAE in the forward pass, so it was not a causal measure	Pre-registered test with 200 controls (raw direction E = 0.0048; SAE directions not distinguishable from random); then **M3**/**M4** patching
L6 chosen on validation; single seed	Causal sites from M3 (L1 slot, L8 final, L11 final); 3 seeds
Frozen probe only	Four readers (M2)
Historical R1 (honesty fine-tune) and historical R2 (feature clamp)	Tested: **both failed**. R1: ReLU inconclusive, TopK/JumpReLU not supported. R2: E ≈ −0.11, below random controls. Both HISTORICAL

D4. Persona information, RepE-LM, VAL (ReLU / TopK / JumpReLU or BatchTopK)

Layer	Metric	Old SAEs	R2	R4 KL	res-jb	Random decoder
8	frozen gap	0.235 / 0.170 / 0.168	0.210 / 0.115 / 0.140	—	0.247	0.202
8	noise gap	0.092 / 0.085 / 0.076	0.067 / 0.054 / 0.052	—	0.075	0.135
8	**re-fit gap**	0.212 / 0.156 / 0.162	0.193 / **0.108** / 0.126	0.106	0.223	0.195
8	SAE − PCA	+0.075 / +0.033 / +0.040	+0.092 / +0.027 / +0.053	—	+0.111	−0.018
11	re-fit gap	0.180 / 0.149 / 0.118	0.194 / 0.109 / 0.101	0.111	0.234	0.154
11	SAE − PCA	+0.104 / +0.083 / +0.052	+0.137 / +0.070 / +0.067	—	+0.167	+0.034

D5. Causal preservation, T_rec/T_full (VAL)

Site	Old SAEs	R2	R4 MSE / KL	Random decoder	DAS rank-4	PCA rank-4
L1 slot	0.938 / 0.931 / 0.968	0.992 / 0.988 / 0.994	—	0.660	1.016	1.000
L8 final	0.621 / 0.625 / 0.665	0.643 / **0.726** / **0.754**	0.720 / 0.710	0.023	0.913	0.518
L11 final	0.677 / 0.620 / 0.678	0.712 / 0.692 / 0.701	0.694 / **0.671**	0.109	0.619	0.587

- **Persona loss is real (unlike GoT):** the frozen gap is about 2× the noise gap.
- **R2 TopK cut the persona re-fit gap by about 30 %,** but not relative to PCA.
- **Causal:** R2 +0.10 at L8 (CI [+0.019, +0.182]); R4 −0.024 at L11 (all seeds).
- **SAE features vs subspaces:** at k = 4, SAE features carry only 8–16 % of the effect.
- **Validation only:** RepE-LM has no test split.

Part E. M5 final test and bottom line

E1. M5 decision summary (TEST)

ID	Comparison	Improved	No clear change	Regressed
C1	old TopK-3072 → R2 TopK-12288	17	7	0
C2	old ReLU → R2 ReLU-12288	17	7	0
C3	old JumpReLU → R2 BatchTopK-12288	17	7	0
A1	data + recipe (width 3,072)	17	7	0
A2	width 3,072 → 12,288	16	6	2 (GoT at L8)
K1	R4 KL vs MSE control	4	2	2 (NMSE)
K4	R4 KL vs R2 base	4	2	2 (NMSE)

Every OWT reconstruction and behaviour cell improved. Every "no clear change" is a GoT cell.

Independent re-check:

- 11/11 code hashes match;
- 81/81 rows, 0 duplicates;
- 136/136 decisions reproduced.

Audit caveats:

- The OWT and GoT test rows had been scored before for the old SAEs ("previously exposed"), though never used for any R2/R4 choice.
- CIs reflect seed variance only.
- There is no multiplicity correction over 136 rows.
- Persona and causal results were not in M5.

E2. Scorecard

Level	Old (corrected baseline)	Now (R2 / R4)	Improved?	Evidence
Reconstruction (OWT)	L8 EV 0.805–0.844	L8 EV 0.868–0.910	**Yes:** NMSE −31 to −74 %	TEST
Behaviour (OWT)	L8 ΔCE 0.121–0.218	R2 0.054–0.131; R4 0.051	**Yes:** −35 to −79 %, then −17 %	TEST
Truth information (GoT)	re-fit gap 0.03–0.11	about the same	**No:** 0/34 cells; never beats PCA	TEST
Persona information (RepE)	re-fit gap 0.12–0.21	0.10–0.19	Absolute only, not relative to PCA	VAL
Causal (RepE)	L8 0.62–0.67	R2 0.73–0.75; R4 L11 −0.024	Mixed	VAL

E3. Failed, unfinished, limitations

Failed	Unfinished	Limitations
• Truth-aware fine-tuning (circular, then worse) • Historical R1/R2 honesty methods • Modern JumpReLU (L0 would not settle) • ReLU/JumpReLU output-KL (L0 drift) • Method A (frozen-probe gain only) • R4 causal effect at L11	• R3 Gemma SAEs • E1 task-L0 matching • RepE-LM test split • Token count vs recipe isolation (10M/25M/50M checkpoints) • L0 frontier; width sweep • Off-machine checkpoint backup	• GPT-2 Small only • 3 seeds • RepE results validation-only • Task-prompt L0 unmatched • Token count and recipe confounded • Test rows previously exposed for the old SAEs

> **Main message.** Better reconstruction improves behavioural faithfulness, but it does not by itself improve truth or causal faithfulness. Each level must be measured separately against noise, PCA and random-decoder references.

Sources (repository-relative):

- Corrected baseline: `results/openwebtext/reconstruction_evaluation/uninterrupted_protocol/`
- R2: `results/openwebtext/r2_scaled_saes/`
- R4: `results/openwebtext/r4_kl_finetune/`
- M2: `results/shared/m2_four_level_faithfulness/`
- M5: `results/shared/m5_confirmatory_test/`
- RepE (M1, M3, M4, R5): `results/repe/repe_lm_gates/`, `results/repe/causal_site_sweep/`, `results/repe/sae_patch_decomposition/`, `results/repe/r5_causal_subspace_skyline/`
- E2: `results/shared/e2_public_sae_reference/`